CAR PARKING MANAGEMENT SYSTEM

by

BHUVISHA GOHIL

150096725190

1. Problem Statement

Parking management has become an important issue in busy places such as shopping malls, offices, universities, and public areas. When parking is managed manually, it can lead to several problems such as difficulty in tracking available parking slots, confusion in assigning slots to vehicles, and errors in calculating parking charges.

Manual systems also make it hard to maintain proper records of vehicles entering and leaving the parking area. This can cause inefficiency and delay in managing parking operations.

To solve these problems, a **Car Parking Management System** is developed using the **C++ programming language**. The system automates the process of parking slot allocation, vehicle tracking, and parking fee calculation.

The system aims to provide a simple and efficient solution that can:

- Automatically allocate available parking slots
- Keep track of vehicles entering the parking area
- Record entry and exit time of vehicles
- Calculate parking fees based on parking duration
- Display the status of parking slots

This system helps in improving parking management and reducing manual work.

2. Solution

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3. Working/Methods & Algorithms.

The Car Parking Management System works using a **menu-driven program**.
When the program starts, the user can choose different operations from the menu.

1. Park Vehicle
- User enters vehicle number and parking hours.
- System finds the first available parking slot.
- Entry time is recorded automatically.
- Parking fee is calculated based on hours.

2. Remove Vehicle
- User enters vehicle number.
- System searches the vehicle in the parking slots.
- Slot is released and parking fee is displayed.
- Exit time is also recorded.

3. View Slots
- Displays all parking slots.
- Shows whether each slot is occupied or available.

4. Search Vehicle
- Finds the slot where a vehicle is parked.

Parking records are also stored in parking_records.txt using file handling.

```
Start
  ↓
Display Menu
  ↓
User Choice
  ↓
 ┌──────────────┬──────────────┬──────────────┬──────────────┐
 │ Park Vehicle │ Remove Vehicle│ View Slots   │ Search Vehicle│
 └──────────────┴──────────────┴──────────────┴──────────────┘
       ↓              ↓              ↓              ↓
  Store Details   Find Vehicle   Show Slot Info   Display Result
       ↓              ↓              ↓              ↓
     Return to Main Menu
            ↓
          Exit
            ↓
          End
```

4. Conclusion

The **Car Parking Management System** helps in managing parking slots in an organized and efficient way.
The system can:

- Automatically allocate parking slots
- Track vehicle entry and exit time
- Calculate parking fees based on parking hours
- Display parking slot availability
- Search for parked vehicles

This project demonstrates the use of **C++ programming, object-oriented concepts, time functions, and file handling** to manage a real-world parking system.

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Thank You